

Appendix D. RF-ADC Electrical Characteristics for the ZU48DR

Reprinted from AMD-Xilinx Document DS926 at <https://docs.xilinx.com/r/en-US/ds926-zynq-ultrascale-plus-rfsoc>

Table 117: RF-ADC Electrical Characteristics for ZU4xDR Devices

Parameter	Comments/Conditions ¹	Min	Typ ²	Max	Units
Analog inputs					
Resolution		14	–	–	Bits
Sample Rate	Devices using quad ADC tile channel	0.5	–	2.5	GS/s
	Devices using dual ADC tile channel	1	–	5	GS/s
Full-scale input ³	Input 100Ω on-die termination when DSA attenuation = 0 dB	–	1	–	V _{PPD}
		–	1	–	dBm
Maximum allowed input power	Input 100Ω on-die termination when DSA attenuation ≥ 15 dB	–	4.8	–	V _{PPD}
		–	14.6	–	dBm
Digital Attenuation Range		0	–	27	dB
Attenuator step size		–	1	–	dB
Auto attenuation	Automatically set when amplitude over-voltage is asserted	–	15	–	dB
Analog input bandwidth ⁴	Full-power bandwidth (–3 dB)	–	6	–	GHz
Return loss (R _L) ⁵	Up to 4 GHz	–	–12	–	dB
	Up to 6 GHz	–	–10	–	dB
Optimized common mode voltage range	Performance optimized range. AC and DC coupling modes ⁶	0.68	0.7	0.72	V
Maximum common mode voltage range	Available range before triggering over-voltage protection. AC and DC coupling modes ⁶	0.4	0.7	1	V
Crosstalk isolation between channels ⁷	F _{IN} = 0–4 GHz	–	–75	–	dBc
	F _{IN} = 0–6 GHz	–	–70	–	dBc

Notes:

1. Analog inputs at –1 dBFS, unless otherwise noted in the test conditions.
2. Typical values are specified at nominal voltage, T_J = 40°C.
3. Full scale range is defined as the approximate input power required to drive the ADC to full scale output for a 5 MHz input tone. The full scale range can vary from dual to quad ADCs. It is also subject to variation across process, voltage, and temperature and from package types. The typical average value is provided.
4. ADC bandwidth is defined as the RF input bandwidth, or where the input amplitude response drops 3 dB relative to a low-frequency reference point of 100 MHz.
5. This is the return loss of the worst case channel quoted from DC to a specified frequency point. Consult the S parameter I/O files for further details because input characteristics depend on channel and package selection. The RL reference plan is close to the BGA footprint, keeping two times the BGA pitch distance from ball contact to avoid micro-probing electromagnetic disturbance.
6. When using DC coupling mode, use the VCM output pin to bias the input to the RF-ADC.
7. Values represent two channel crosstalk worst-case values for any combination of channel selections. This specification is only characterized on the XCZU46DR-H1760.

Appendix E. RF-DAC Electrical Characteristics for the ZU48DR

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Table 132: RF-DAC Electrical Characteristics for ZU4xDR Devices

Parameter	Comments/Conditions ¹	Min	Typ ²	Max	Units
Analog Outputs					
Resolution		14	–	–	Bits
Sample rate ³	-2E, -2I, -2LI speed grade without clock forwarding, datapath modes 2, 3, and 4	0.5	–	9.85	GS/s
	-2E, -2I, -2LI speed grade with clock forwarding, datapath modes 2, 3, and 4	0.5	–	9.70	GS/s
	-1E, -1I, -1LI, -1M speed grade, datapath modes 2, 3, and 4	0.5	–	8.92	GS/s
	All speed grades, datapath mode 1	0.5	–	7.0	GS/s
Maximum output power	$V_{DAC_AVTT} = 3.0V$, 100 Ω termination, signal frequency <200 MHz	–18.5	–	6.5	dBm
Output current range	AC coupling: $V_{DAC_AVTT} = 3.0V$, 100 Ω termination, signal frequency <200 MHz	2.25	–	40.5	mA
	DC coupling: $V_{DAC_AVTT} = 3.0V$, 100 Ω termination, signal frequency <200 MHz	6.4	–	32	mA
Variable output current step size		–	43.75	–	μA
Variable output power range Equivalent dynamic range from output current range ⁴	At 240 MHz	24	–	–	dB
	At 3500 MHz	20	–	–	dB
	At 4900 MHz	18	–	–	dB
	At 5900 MHz	17	–	–	dB
Analog bandwidth	Full power bandwidth (–3 dB)	–	6	–	GHz
Return loss (R_L) ⁵	Up to 4 GHz	–	–12	–	dB
	Up to 6 GHz	–	–10	–	dB
On-die termination	Single-ended on-die termination to external 3V V_{DAC_AVTT}	–	50	–	Ω
Crosstalk isolation between channels ⁶	$F_{OUT} = 0\text{--}4$ GHz	–	–75	–	dBc
	$F_{OUT} = 0\text{--}6$ GHz	–	–70	–	dBc

Notes:

1. RF-DAC sampling rate is the highest rate using external sampling clock.
2. Typical values are specified at nominal voltage, $T_j = 40^\circ C$.
3. See the *Zynq UltraScale+ RFSoc RF Data Converter LogiCORE IP Product Guide (PG269)* for additional information on datapath modes.
4. The variable output power effective dynamic range depends on the signal frequency that is shown in the equivalent dynamic range. The specification is supplied for AC coupling mode. A derating of 12 dB is applied for DC coupling mode.
5. This is the return loss of the worst case channel quoted from DC to a specified frequency point. Consult the S parameter I/O files for further details as input characteristics depend on channel and package selection. The RL reference plan is closed to the BGA footprint, keeping two times the BGA pitch distance from ball contact to avoid micro-probing electromagnetic disturbance.
6. Values represent two channel crosstalk worst-case values for any combination of channel selections. This specification is only characterized on XCZU46DR-H1760.